

EE446 Lab 9
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The primary software fix was in the model compilation section. The original used `loss = 'mse'` and `metrics = 'mse'` which are meant for regression objects. I changed them to use `model.compile(optimizer='rmsprop', loss='categorical_crossentropy', metrics=['accuracy'])`

These are much better suited for a softmax classifier because it avoids gradient vanishing which leads to early plateauing in the learning phase. Categorical cross-entropy enables stronger gradients when predictions are incorrect. There were other downstream changes in the code but all stemmed from this key debugging fix.

Below the screenshots show evidence of the classifier finding both ASL signs.



Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem1101')



```
-----  
DEBUG: Motion detected! Starting sample collection...  
DEBUG: Waiting for acceleration data...  
DEBUG: Waiting for acceleration data...  
DEBUG: Waiting for acceleration data...  
DEBUG: Waiting for acceleration data...  
DEBUG: Collected 20 / 119  
DEBUG: Collected 40 / 119  
DEBUG: Collected 60 / 119  
DEBUG: Collected 80 / 119  
DEBUG: Collected 100 / 119  
DEBUG: All samples collected! Running inference...  
DEBUG: Inference complete! Results:  
hi: 0.067791  
sup: 0.932209
```

```
DEBUG: accelerationAvailable = 1  
DEBUG: aX=0.48 aY=-1.59 aZ=0.63 sum=2.69  
DEBUG: Motion detected! Starting sample collection...  
DEBUG: Waiting for acceleration data...  
DEBUG: Waiting for acceleration data...  
DEBUG: Collected 20 / 119  
DEBUG: Collected 40 / 119  
DEBUG: Collected 60 / 119  
DEBUG: Collected 80 / 119  
DEBUG: Collected 100 / 119  
DEBUG: All samples collected! Running inference...  
DEBUG: Inference complete! Results:  
hi: 0.233360  
sup: 0.766640
```

```
DEBUG: accelerationAvailable = 1  
DEBUG: aX=-0.04 aY=0.15 aZ=1.32 sum=1.51  
DEBUG: aX=-0.20 aY=0.23 aZ=1.20 sum=1.63  
DEBUG: aX=0.31 aY=0.52 aZ=0.88 sum=1.71  
DEBUG: Motion detected! Starting sample collection...  
DEBUG: Waiting for acceleration data...  
DEBUG: Waiting for acceleration data...  
DEBUG: Waiting for acceleration data...  
DEBUG: Waiting for acceleration data...  
DEBUG: Collected 20 / 119  
DEBUG: Collected 40 / 119  
DEBUG: Collected 60 / 119  
DEBUG: Collected 80 / 119  
DEBUG: Collected 100 / 119  
DEBUG: All samples collected! Running inference...  
DEBUG: Inference complete! Results:  
hi: 0.366901  
sup: 0.633099
```

```
DEBUG: accelerationAvailable = 1  
DEBUG: Motion detected! Starting sample collection...  
DEBUG: Waiting for acceleration data...  
DEBUG: Collected 20 / 119  
DEBUG: Collected 40 / 119  
DEBUG: Collected 60 / 119  
DEBUG: Collected 80 / 119  
DEBUG: Collected 100 / 119  
DEBUG: All samples collected! Running inference...  
DEBUG: Inference complete! Results:  
hi: 0.811150  
sup: 0.188850
```



```
DEBUG: accelerationAvailable = 1
```



Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem1101')



```
DEBUG: Collected 20 / 119
DEBUG: Collected 40 / 119
DEBUG: Collected 60 / 119
DEBUG: Collected 80 / 119
DEBUG: Collected 100 / 119
DEBUG: All samples collected! Running inference...
DEBUG: Inference complete! Results:
hi: 0.510772
sup: 0.489228

DEBUG: accelerationAvailable = 1
DEBUG: Motion detected! Starting sample collection...
DEBUG: Waiting for acceleration data...
DEBUG: Collected 20 / 119
DEBUG: Collected 40 / 119
DEBUG: Collected 60 / 119
DEBUG: Collected 80 / 119
DEBUG: Collected 100 / 119
DEBUG: All samples collected! Running inference...
DEBUG: Inference complete! Results:
hi: 0.059743
sup: 0.940257

DEBUG: accelerationAvailable = 1
DEBUG: Motion detected! Starting sample collection...
DEBUG: Waiting for acceleration data...
DEBUG: Collected 20 / 119
DEBUG: Collected 40 / 119
DEBUG: Collected 60 / 119
DEBUG: Collected 80 / 119
DEBUG: Collected 100 / 119
DEBUG: All samples collected! Running inference...
DEBUG: Inference complete! Results:
hi: 0.042225
sup: 0.957775

DEBUG: accelerationAvailable = 1
DEBUG: aX=-0.18 aY=0.73 aZ=-0.48 sum=1.40
DEBUG: aX=-0.49 aY=0.26 aZ=-0.52 sum=1.26
DEBUG: Motion detected! Starting sample collection...
DEBUG: Waiting for acceleration data...
DEBUG: Waiting for acceleration data...
DEBUG: Waiting for acceleration data...
DEBUG: Waiting for acceleration data...
DEBUG: Collected 20 / 119
DEBUG: Collected 40 / 119
DEBUG: Collected 60 / 119
DEBUG: Collected 80 / 119
DEBUG: Collected 100 / 119
DEBUG: All samples collected! Running inference...
DEBUG: Inference complete! Results:
hi: 0.300553
sup: 0.699447

DEBUG: accelerationAvailable = 1
DEBUG: Motion detected! Starting sample collection...
DEBUG: Waiting for acceleration data...
DEBUG: Collected 20 / 119
DEBUG: Collected 40 / 119
DEBUG: Collected 60 / 119
DEBUG: Collected 80 / 119
DEBUG: Collected 100 / 119
DEBUG: All samples collected! Running inference...
```

